CLINIC APPOINTMENT MANAGER

A PROJECT REPORT

Submitted by

SANGAMITHRA ARUN

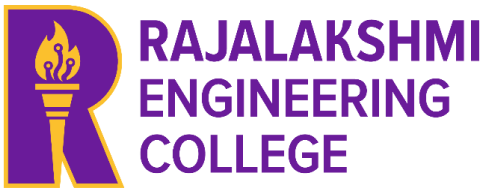
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CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING
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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "CLINIC APPOINTMENT MANAGER" is the bonafide work of **SANGAMITHRA ARUN (240701465)**, who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, SANGAMITHRA ARUN (240701465), hereby declare that the project report titled **"CLINIC APPOINTMENT MANAGER"**, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report – including its data model, forms, reports, pages, Deluge scripts and workflows – was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature, code snippet or external resource used in this work has been duly acknowledged in the report.

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Date: _____

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SANGAMITHRA ARUN

ABSTRACT

Small and mid-sized clinics still manage appointments through paper registers and spreadsheets. Staff quote consultation fees by hand, update statuses inconsistently, and cannot see in real time how many appointments are scheduled, completed or cancelled, or which specializations and time slots are busiest. Patients, in turn, must phone the clinic to confirm their bookings.

This project presents the Clinic Appointment Manager, a low-code application built on Zoho Creator, whose visual builders let the whole solution be delivered quickly and maintained by a non-programmer. The application contains three linked forms (Appointment Record, Doctor Profile and Patient Profile), seven reports, and a dashboard page with four KPI cards, a status summary, two bar charts and quick links. Three Deluge automations set the default status to Scheduled, look up the consultation fee from the selected doctor, and stamp the completion date when an appointment is completed. Role-based access separates administrators, staff and patients, and patients use a portal with a filtered report that shows only their own appointments.

Testing against live data confirmed that the dashboard counts (4 doctors, 3 patients, 2 appointments, $55.00 in consultation fees) match the reports, that completion dates are populated automatically, and that cancelled appointments remain available for audit. The application removes manual fee entry and manual counting, and gives the clinic a real-time view of its workload.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Role-Based Access Control; Clinic Appointment Management*

PROJECT AT A GLANCE

Attribute	Details
Project Title	Clinic Appointment Manager
Application Domain	Healthcare (clinic appointment management)
Student Name & Register No.	SANGAMITHRA ARUN – 240701465
Zoho Creator Edition	[Education / Free / Professional]
Development Platform	Zoho Creator low-code platform
Number of Forms	3
Number of Reports (by type)	7 (List: 4, Grouped: 3)
Number of Pages / Dashboards	1 dashboard page
Number of Workflows	Form: 3 | Schedule: 0 | Approval: 0 | Blueprint: 0 | Batch: 0
Deluge Functions / Custom Actions	3 form-event scripts
Integrations	None in this release
Roles & Profiles Defined	Administrator, Staff, Patient (Portal)
Environments Used	[Development → Production]

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FR / NFR	Functional / Non-Functional Requirement
KPI	Key Performance Indicator
LCAP	Low-Code Application Platform
RBAC	Role-Based Access Control
REST	Representational State Transfer
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Outpatient clinics coordinate three kinds of information every day: who the patients are, which doctors (and specializations) are available, and when each consultation takes place. In a multi-specialist clinic offering Orthopedics, Dermatology and Cardiology, this coordination is performed by front-desk staff who book a time slot, quote the consultation fee, and later record whether the visit was completed or cancelled. In many small clinics this still happens through paper registers, phone calls and spreadsheets.

A typical day at such a clinic starts with the receptionist reviewing the doctors available for each specialization and the time slots still open. Patients are then booked one by one: the receptionist records the patient's name, the chosen doctor, the date and time slot, and the reason for the visit, and tells the patient the consultation fee. After the consultation, the entry has to be marked as completed, or as cancelled if the patient did not come. At the end of the day or week, the clinic owner wants to know how many appointments were scheduled, completed and cancelled, which doctors and specializations were in demand, and how much consultation revenue was generated.

The Clinic Appointment Manager developed in this project digitises that process on the Zoho Creator low-code platform. The live application currently holds 4 doctor profiles, 3 patient profiles and 2 appointment records, and is operated by clinic staff, with patients viewing their own appointments through a portal.

1.2 Problem Statement

Clinic staff need a single, reliable place to register doctors and patients, book appointments, track their status and see the day's workload. Without it, the following pain points arise:

- Appointment details, fees and statuses are scattered across registers and spreadsheets, causing duplication and errors.
- Consultation fees are entered manually and can differ from the doctor's actual fee.
- Completion and cancellation dates are not recorded consistently, so history cannot be audited.
- Management has no real-time view of appointment counts, status mix, time-slot load or specialization demand.
- Patients cannot view their own appointments without calling the clinic.

These problems are common to small healthcare providers that have no budget for a full hospital information system but still need dependable records. A lightweight, configurable application that staff can learn in minutes and an administrator can modify without a programmer is therefore the right level of solution.

1.3 Objectives

- To design a normalised data model and forms for doctors, patients and appointments.
- To automate default status, fee lookup and completion-date capture using Deluge scripting.
- To provide a dashboard and grouped reports for monitoring appointments, statuses and specializations.
- To provide patients with a secure portal view restricted to their own appointments.
- To deploy the application with role-based access and governance controls.

1.4 Scope of the Project

In scope: three forms (Appointment Record, Doctor Profile, Patient Profile); seven reports; one analytics dashboard; three Deluge automations; role-based access for administrators, clinic staff and patient portal users; web access. **Out of scope:** billing and payment collection, electronic medical records, SMS or e-mail gateways, and a native mobile build.

1.5 Justification for a Low-Code Approach

Low-code platforms let small teams deliver data-centric business applications quickly, with hosting, security and mobile access provided by the platform [1], [5]. For a clinic scheduler, whose logic is mainly forms, lookups, status rules and reports, Zoho Creator provides each of these natively [2], while Deluge [3] covers the few rules that need scripting. The result is faster delivery and an application that a non-programmer administrator can maintain.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing system, a patient phones or walks in; the receptionist checks the doctor's availability in a register, writes the slot, quotes a fee from memory or a price list, and later updates the entry by hand when the visit is completed or cancelled. Statistics are produced by counting entries manually (Figure 2.1).

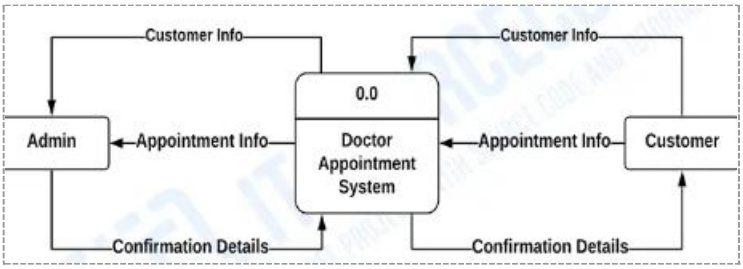


Figure 2.1: As-Is Process Flow of the Existing System

This manual flow also depends on the individual receptionist's memory and handwriting. When a different person covers the desk, doctor availability, fee changes and cancellations are easily missed, and there is no single source of truth that the owner can consult.

2.2 Limitations of the Existing System

- No validation: double bookings and wrong fees go unnoticed.
- Status and completion dates are updated inconsistently or not at all.
- Reporting by status, specialization or time slot requires manual counting.
- No role-based access; patients cannot self-serve.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Several weeks for forms, back end and dashboard [estimate: 6–8 weeks]	Building blocks configured directly [estimate: 1–2 weeks]
Skills required	Front end, back end, database, DevOps	Form design and basic Deluge
Hosting & infrastructure	Server, database and deployment set up by the developer	Provided by the platform
Mobile application	Separate build required	Zoho Creator mobile app / portal
Security & access control	Custom authentication and roles	Built-in roles, profiles and portal
Maintenance & change	Code changes, redeployment	Edit forms and scripts in the builder
Customisation limits	Unlimited	Bound by platform features and Deluge limits
Cost	Developer time and hosting	Platform subscription (education edition)

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Source & Year	Title / Product	Key Findings	Relevance
1	Bock & Frank, 2021 [6]	Low-Code Platform	Defines low-code platforms and their model-driven, visual development approach	Positions Zoho Creator as an LCAP
2	Luo et al., 2021 [7]	Characteristics and Challenges of Low-Code Development	Practitioners value speed but report platform limits and learning-curve issues	Informs Section 8.4 limitations
3	Sahay et al., 2020 [8]	Supporting the Understanding and Comparison of Low-Code Development Platforms	Proposes criteria for comparing platforms	Supports platform choice
4	Waszkowski, 2019 [9]	Low-Code Platform for Automating Business Processes in Manufacturing	Low-code speeds up process automation	Similar automation of a business workflow
5	Zoho Corp. [2], [3]	Zoho Creator and Deluge documentation	Forms, reports, pages, workflows and Deluge scripting	Direct implementation reference

2.5 Emerging Trends in Low-Code Development

Current trends include AI-assisted app building, citizen development and hyper-automation. These influenced the design: the application is kept simple enough for a citizen developer to maintain, automation is limited to rules that remove repeated manual steps, and AI features were considered but not added because the data volume is small (Section 5.9).

Citizen development in particular matters here: the clinic administrator can extend the application, for example by adding a new specialization or report, without writing code. At the same time, Deluge gives the developer enough control to enforce rules such as default status and fee lookup, so the platform covers the whole range from configuration to scripting.

2.6 Proposed System

Table 2.3: Limitations Addressed by the Proposed System

Limitation of Existing System	How the Application Addresses It
No validation of bookings or fees	Lookups to doctor and patient records; fee fetched automatically from the doctor
Inconsistent status and completion dates	Default status Scheduled; Completion Date stamped automatically
Manual counting for reports	Dashboard KPI cards, charts and grouped reports
No role-based access or self-service	Roles, permission profiles and a filtered patient portal report

The proposed Zoho Creator application replaces the register with three linked forms, defaults and auto-calculated fields that remove manual entry, grouped reports and a live dashboard that remove manual counting, and a filtered portal report that gives patients self-service access to their own appointments.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Clinic owner / system manager	Manage users and settings; maintain doctor profiles; view all data and the dashboard
Staff (Scheduler)	Front-desk employee (e.g., user "san")	Register patients; book and update appointments
Patient (Portal User)	Registered patient	View own appointments through My Appointments (Patient)

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	Register a doctor with name and specialization	Admin	High
FR-02	Register a patient with name and gender	Staff	High
FR-03	Book an appointment by selecting patient, doctor, date, time slot and reason for visit	Staff	High
FR-04	Set status to Scheduled automatically on a new appointment	System	High
FR-05	Fetch the consultation fee automatically from the selected doctor	System	High
FR-06	Record Completion Date automatically when status becomes Completed	System	High
FR-07	Retain cancelled appointments for audit and history	System	Medium
FR-08	Show a dashboard with KPI cards, status summary and charts	Admin, Staff	Medium
FR-09	Provide reports grouped by status, specialization and gender	Admin, Staff	Medium
FR-10	Show a patient only their own appointments through the portal	Patient	High
FR-11	Provide quick links to add records and open reports from the dashboard	Admin, Staff	Low

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports and the dashboard shall load within a few seconds for clinic-scale data
NFR-02	Security	Users shall access only the data permitted by their role; patients only their own records
NFR-03	Usability	Forms shall use dropdowns and lookups to minimise typing; usable on a standard web browser
NFR-04	Availability	Hosted on the Zoho Creator cloud; no separate server to maintain
NFR-05	Maintainability	Changes shall be made in the form, report and Deluge editors without redeployment
NFR-06	Auditability	Cancelled records remain in the database; record changes are traceable

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator [Education / Free / Professional] Edition
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)

Category	Specification
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

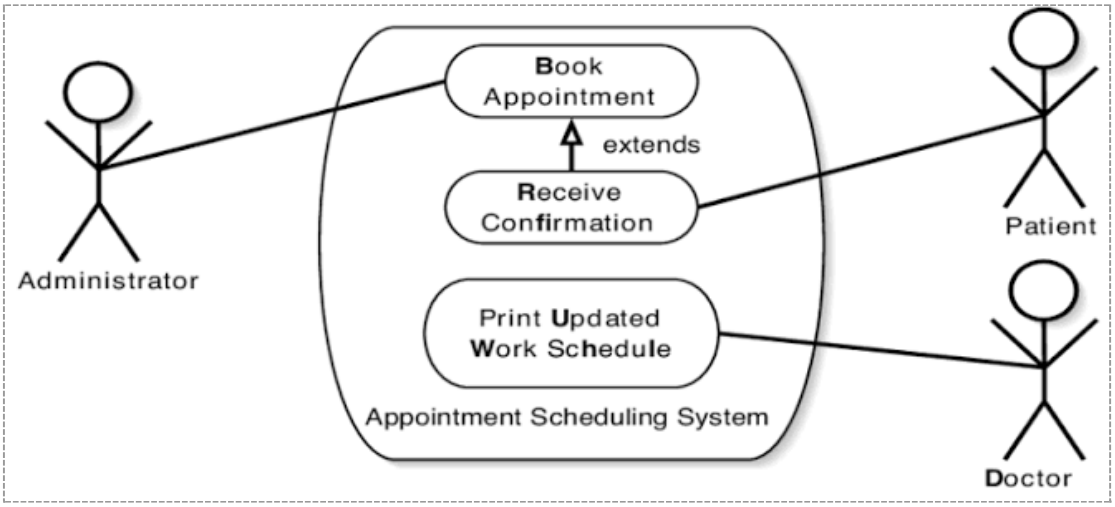


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Book an Appointment

Use Case ID	UC-01
Actor(s)	Staff (Scheduler)
Pre-condition	Doctor and patient profiles already exist
Main Flow	1. Staff opens + Appointment Record. 2. Selects patient and doctor; fee is filled automatically. 3. Enters date, time slot and reason. 4. Status defaults to Scheduled. 5. Submits.
Alternate Flow	If a mandatory field is empty, the form shows a validation message and the record is not saved.
Post-condition	Appointment appears in All Appointment Records and in the dashboard counts
Related FR	FR-03, FR-04, FR-05

Table 3.6: Use Case Description – Complete or Cancel an Appointment

Use Case ID	UC-02
Actor(s)	Staff (Scheduler)
Pre-condition	An appointment exists with status Scheduled
Main Flow	1. Staff opens All Appointment Records. 2. Opens the record in edit mode. 3. Changes Status to Completed or Cancelled. 4. Submits the update. 5. For Completed, the Completion Date is filled automatically.
Alternate Flow	For Cancelled, no completion date is stored and the record is kept unchanged for audit.
Post-condition	Dashboard status summary and Statuses report reflect the new status
Related FR	FR-06, FR-07, FR-08

Table 3.7: Use Case Description – Patient Views Own Appointments

Use Case ID	UC-03
Actor(s)	Patient (Portal User)
Pre-condition	Patient has portal access and at least one appointment
Main Flow	1. Patient signs in to the portal. 2. Opens My Appointments (Patient). 3. Views date, time slot, doctor, reason and status of own appointments.
Alternate Flow	If the patient has no appointments, the report is empty.
Post-condition	No other patient's data is visible
Related FR	FR-10

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

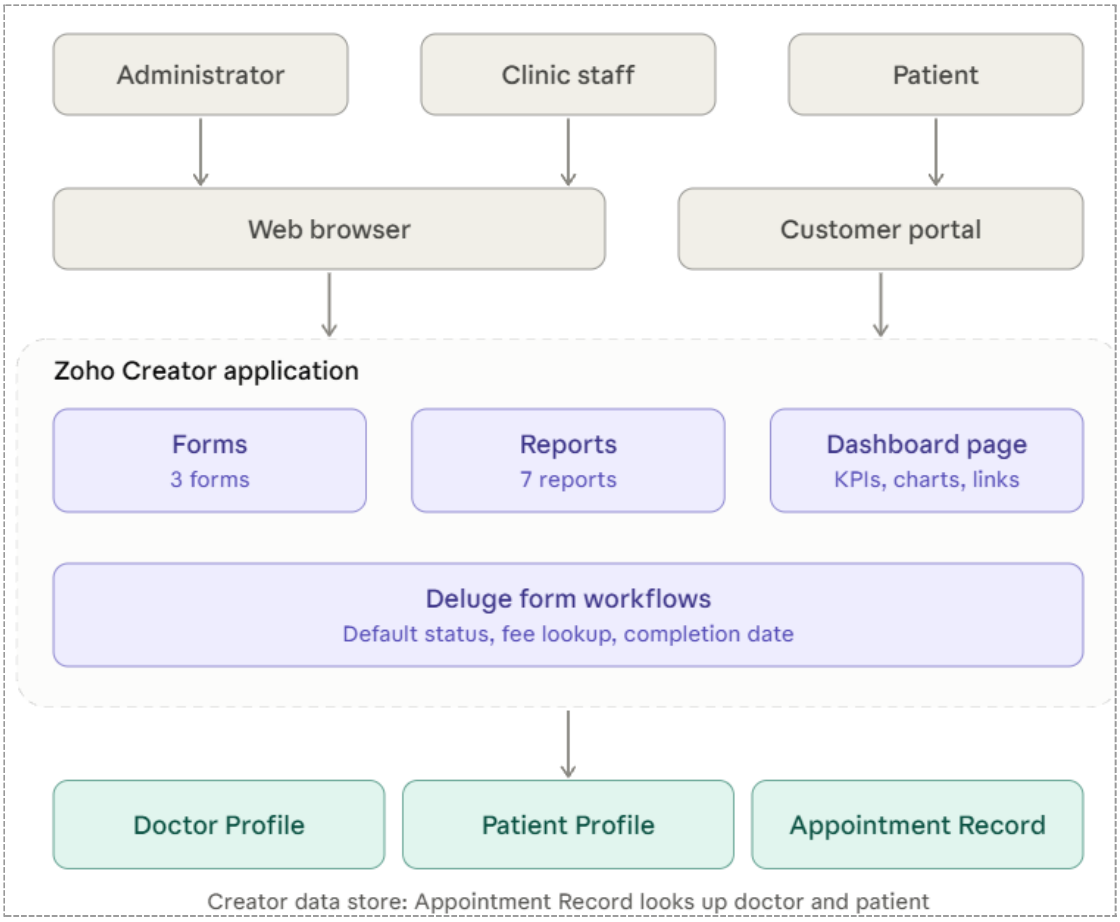


Figure 4.1: System Architecture of the Application

The architecture has four layers. The **access layer** consists of the web browser for staff and administrators and the customer portal for patients. The **application layer** is the Zoho Creator app itself, with the navigation menu, forms, reports and the dashboard page. The **logic layer** contains the Deluge form workflows that set defaults, look up fees and stamp completion dates. The **data layer** is the Creator database that stores the Doctor Profile, Patient Profile and Appointment Record forms and the lookups between them.

Users (administrator, staff, patient) reach the application through the web browser or the customer portal. The Zoho Creator layer contains the three forms, seven reports, the dashboard page and the Deluge form workflows. Data is stored in the Creator data layer. No external integration is used in this release.

4.2 Data Model

Appointment Record is the central entity; each appointment refers to exactly one patient and one doctor, while a doctor or patient can have many appointments.

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship	Implemented Using
Doctor Profile	Doctor name, Specialization, Consultation fee	Appointment Record	One-to-Many	Lookup in Appointment
Patient Profile	Patient name, Gender	Appointment Record	One-to-Many	Lookup in Appointment
Appointment Record	Patient, Doctor, Date, Time Slot, Status, Fee	Doctor, Patient	Many-to-One	Two Lookup fields

4.3 Form Design

Table 4.2: Field Specification – Appointment Record Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default
Patient	Patient	Lookup (Patient Profile)	Yes	Select existing patient
Doctor	Doctor	Lookup (Doctor Profile)	Yes	Triggers fee lookup
Appointment Date	Appointment_Date	Date	Yes	—
Time Slot	Time_Slot	Drop Down	Yes	e.g., 10:30, 11:00
Reason For Visit	Reason_For_Visit	Multi Line	Yes	—
Status	Status	Drop Down	Yes	Default: Scheduled
Consultation Fee	Consultation_Fee	Currency	No	Auto from doctor
Scheduled By	Scheduled_By	Single Line	No	Logged-in user [verify]
Completion Date	Completion_Date	Date	No	Auto on Completed
Reminder Sent	Reminder_Sent	Decision Box	No	Default: unchecked

Table 4.3: Field Specification – Doctor Profile and Patient Profile Forms

Form	Field Label	Field Type	Mandatory	Remarks
Doctor Profile	Doctor Name	Name / Single Line	Yes	
Doctor Profile	Specialization	Drop Down	Yes	Orthopedics, Dermatology, Cardiology…
Doctor Profile	Consultation Fee	Currency	Yes	Source of fee lookup [verify]
Patient Profile	Patient Name	Name / Single Line	Yes	
Patient Profile	Gender	Radio / Drop Down	Yes	Used by Genders report

Every form uses mandatory flags on its principal fields and drop-downs for fixed values such as Status and Time Slot, so that data is entered consistently. Lookup fields are used instead of free text for patient and doctor so that names are never mistyped and so that reports can group and filter on them.

4.4 Process and Workflow Design

When a new appointment form loads, Status is set to Scheduled. On selecting a doctor, the fee is fetched. When Status is changed to Completed, the Completion Date is stamped with the current date. A Cancelled record is kept unchanged for history.

4.5 User Interface and Navigation Design

The application has four navigation tabs: **Dashboard**; **Appointment Records** (+ Appointment Record, All Appointment Records, Statuses, My Appointments (Patient)); **Doctor Profiles** (+ Doctor Profile, All Doctor Profiles, Specializations); and **Patient Profiles** (+ Patient Profile, All Patient Profiles, Genders).

4.6 Access Control Design

Table 4.4: Role–Permission Matrix (A = Add, E = Edit, V = View, D = Delete)

Component	Administrator	Staff	Patient (Portal)
Form: Appointment Record	A E V D	A E V	—
Form: Doctor Profile	A E V D	V	—
Form: Patient Profile	A E V D	A E V	—
Report: All Appointment Records	V	V	—
Report: My Appointments (Patient)	V	V	V (own)
Page: Dashboard	V	V	—

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application was created in the Zoho Creator workspace and organised into four sections: Dashboard, Appointment Records, Doctor Profiles and Patient Profiles. Each section groups the "add" form with its reports, so that users reach every function from one menu (Figure 5.1).

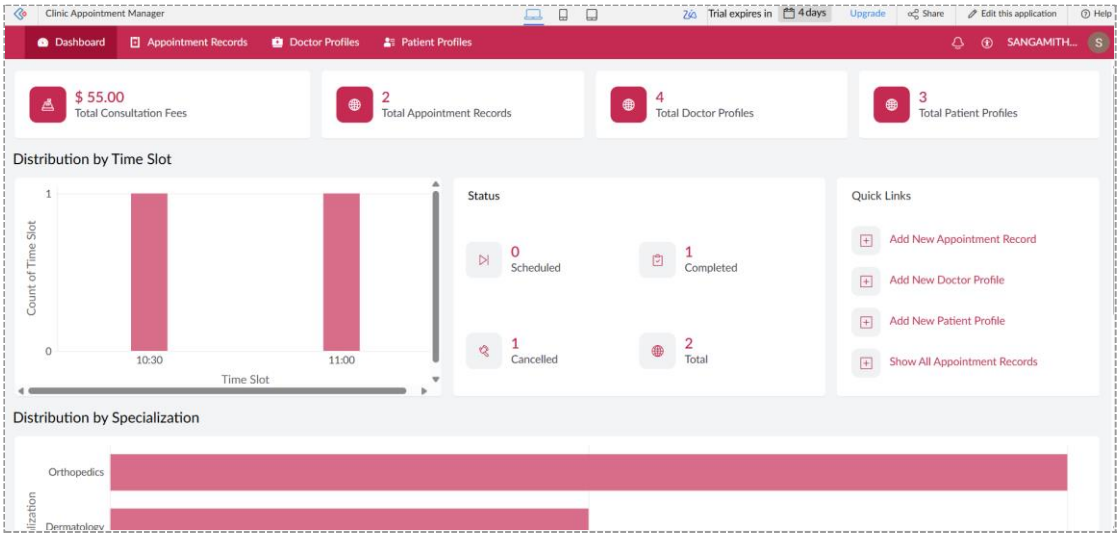


Figure 5.1: Application Home and Navigation Sections

Table 5.1: Application Navigation Structure

Section	Components	Purpose
Dashboard	Dashboard page	Real-time analytics and quick links
Appointment Records	+ Appointment Record, All Appointment Records, Statuses, My Appointments (Patient)	Book, track and group appointments; patient portal view
Doctor Profiles	+ Doctor Profile, All Doctor Profiles, Specializations	Register and categorise doctors
Patient Profiles	+ Patient Profile, All Patient Profiles, Genders	Register and categorise patients

5.2 Forms and Data Capture

Three forms capture the data. Appointment Record uses Lookup fields to Doctor Profile and Patient Profile, a Drop Down for Time Slot and Status, a Date field for Appointment Date and Completion Date, and a Decision Box for Reminder Sent. Mandatory flags prevent incomplete bookings. Figures 5.2 and 5.3 show the form in the builder and in live mode.

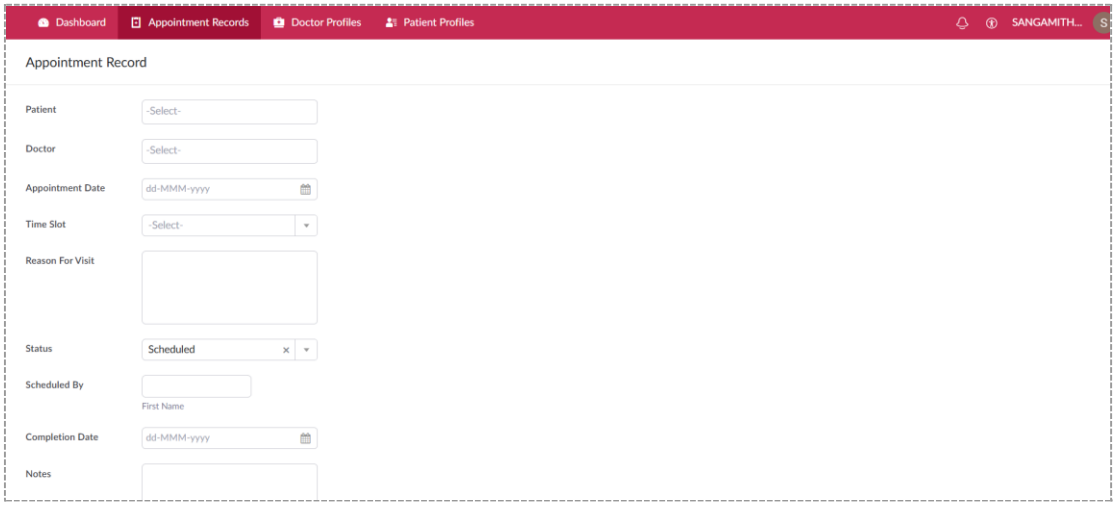


Figure 5.3: Appointment Record Form in Live Mode

The Doctor Profile form registers each doctor once, so the doctor's name, specialization and fee are reused in every appointment through the lookup. The Patient Profile form likewise stores each patient once with the gender used by the Genders report (Figures 5.4 and 5.5).

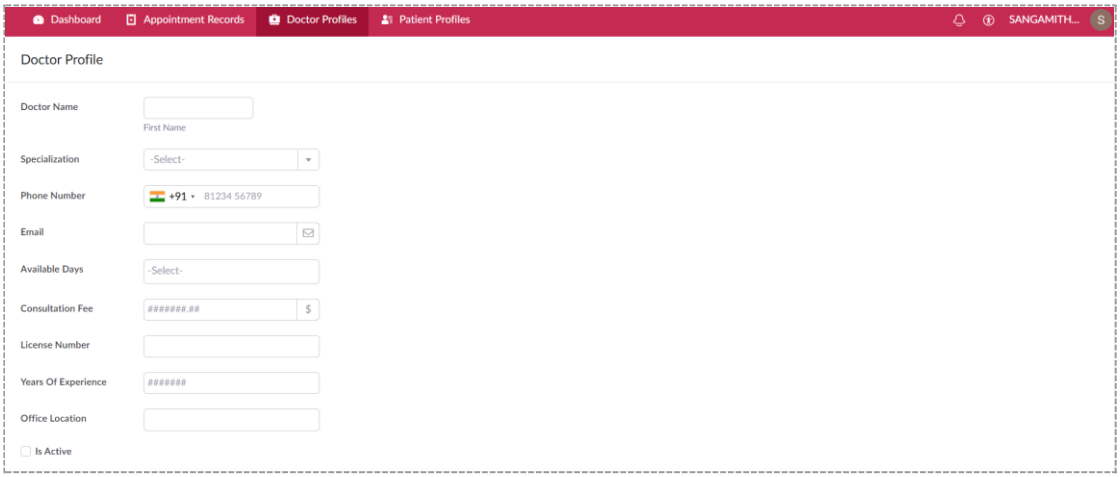


Figure 5.4: Doctor Profile Form in Live Mode

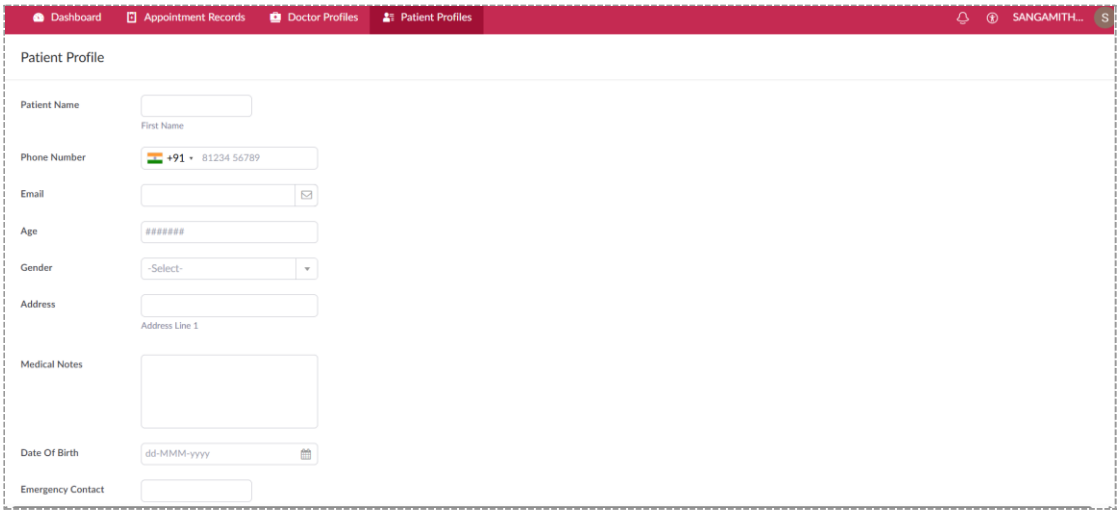


Figure 5.5: Patient Profile Form in Live Mode

5.3 Reports

Table 5.2: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping	Roles
All Appointment Records	List	Appointment Record	Ungrouped; all records	Admin, Staff
Statuses	Grouped list	Appointment Record	Grouped by Status	Admin, Staff
My Appointments (Patient)	Filtered list	Appointment Record	Records of the logged-in patient only	Patient portal
All Doctor Profiles	List	Doctor Profile	All doctors	Admin, Staff
Specializations	Grouped list	Doctor Profile	Grouped by Specialization	Admin, Staff
All Patient Profiles	List	Patient Profile	All patients	Admin, Staff
Genders	Grouped list	Patient Profile	Grouped by Gender	Admin, Staff

The live data confirms the reports work: All Appointment Records lists two appointments (patient joseph with Dr. Yashu on 14-Oct-2026, 11:00, Cancelled; patient ramya with Dr. Priya on 29-Oct-2026, 10:30, Completed).

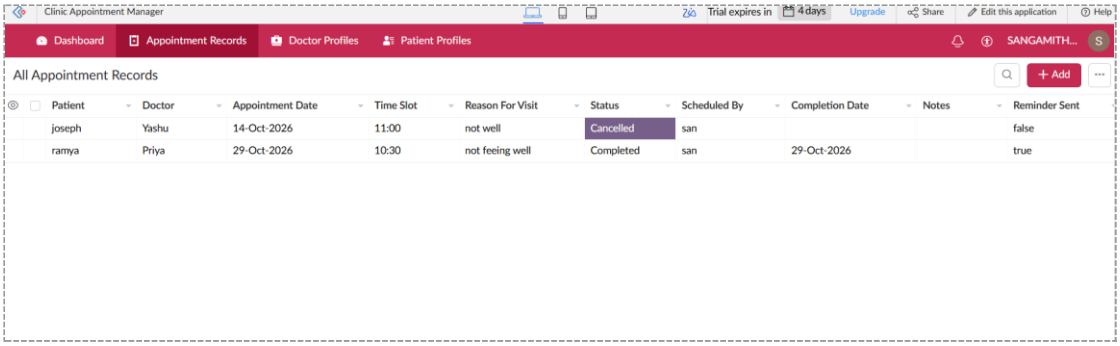


Figure 5.6: All Appointment Records Report

The Statuses report groups the same records by Status, My Appointments (Patient) is the filtered portal view, and Specializations groups doctors by specialization (Figures 5.7 to 5.9).

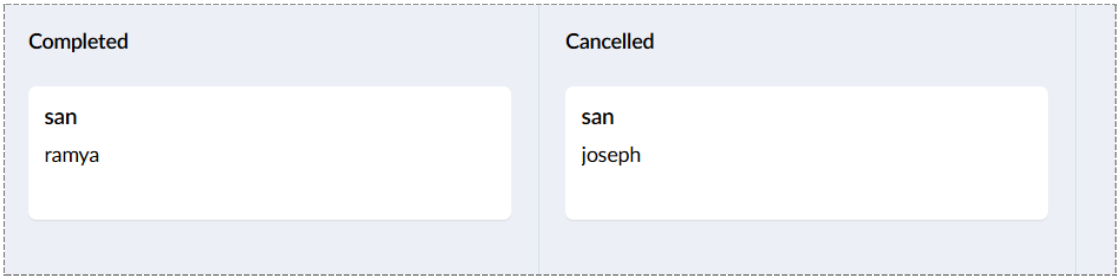


Figure 5.7: Statuses Report

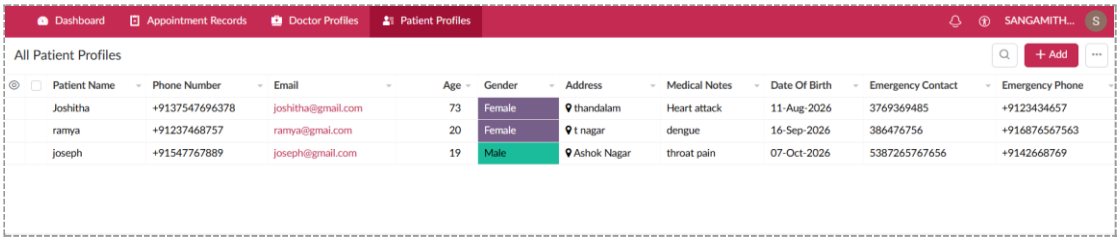


Figure 5.8: My Appointments (Patient) Report

5.4 Pages and Dashboards

The Dashboard is a Creator page built in the UI Builder for administrators and staff. It contains: **four KPI cards** – Total Consultation Fees ($55.00), Total Appointment Records (2), Total Doctor Profiles (4) and Total Patient Profiles (3); a **status distribution summary** – 0 Scheduled, 1 Completed, 1 Cancelled, 2 Total; a **bar chart** of appointments by time slot (10:30 and 11:00); a **horizontal bar chart** by specialization (Orthopedics, Dermatology, Cardiology); and a **Quick Links panel** with buttons to add an appointment, doctor or patient and to show all appointment records.

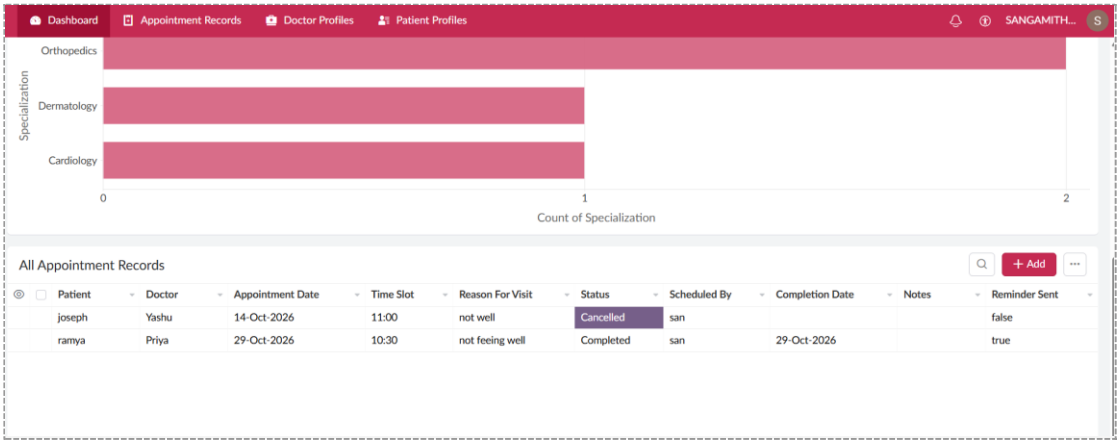


Figure 5.10: Clinic Appointment Manager Dashboard

The KPI cards and status summary give an at-a-glance view of the clinic, while the two bar charts show load by time slot and by specialization (Figures 5.11 and 5.12).

5.5 Deluge Scripting

Table 5.3: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Appointment Record → On Load (add)	Default Status to Scheduled	input field assignment
DS-02	Appointment Record → On User Input of Doctor	Fetch fee from selected doctor	fetch by ID, null check
DS-03	Appointment Record → On Edit (validate / success)	Stamp Completion Date when Completed	if-condition, zoho.currentdate

DS-01 runs when the Appointment Record form loads and assigns the value Scheduled to Status, so staff never have to choose it and a new booking is always in a known state. **DS-02** runs when the Doctor field changes; it fetches the selected doctor's record using its ID and copies the fee into the Consultation Fee field. **DS-03** runs when a record is edited and saved; if Status is Completed and Completion Date is still empty, it writes today's date. The empty-value check prevents an existing completion date from being overwritten.

Code 5.1: DS-02 – dynamic fee lookup (see Appendix C for all scripts)

```
// DS-02 | Form: Appointment_Record | Field: Doctor | On User Input
if(input.Doctor != null)
{
    doc = Doctor_Profile[ID == input.Doctor];
    input.Consultation_Fee = doc.Consultation_Fee;
}
```

Code 5.2: DS-03 – automatic completion date

```
// DS-03 | Form: Appointment_Record | Event: On Edit - On Validate
if(input.Status == "Completed" && input.Completion_Date == null)
{
    input.Completion_Date = zoho.currentdate;
}
```

The result is visible in the data: the Completed appointment for ramya carries Completion Date 29-Oct-2026 without manual entry, while the Cancelled record for joseph keeps a blank completion date.

5.6 Workflows and Automation

Table 5.4: Workflows Implemented

Workflow Name	Type	Trigger / Event	Action	Script
Default Status	Form workflow	Appointment form On Load	Set Status = Scheduled	DS-01
Dynamic Fee Lookup	Form workflow	On user input of Doctor	Fill Consultation Fee	DS-02
Auto Completion Date	Form workflow	On edit when Status = Completed	Set Completion Date	DS-03

Schedule, approval, Blueprint and batch workflows were not used. The booking process has a single step with no multi-level approval, and no recurring job was required; a daily reminder schedule is listed as a future enhancement.

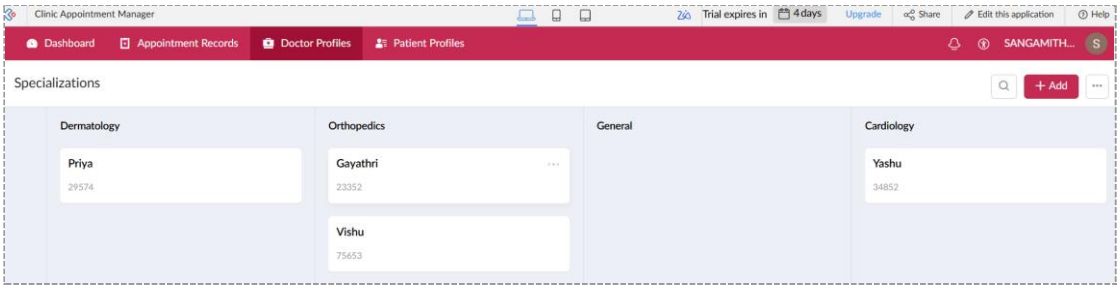


Figure 5.13: Form Workflow Configuration

5.7 Integrations

No external integration or REST API call was used in this release because the clinic's data lives entirely inside the application. Creator's connections, invokeurl and Zoho Mail / CRM integrations can later add e-mail reminders or billing.

5.8 User Management and Role-Based Access

Users are assigned to roles and permission profiles following Table 4.5. Patients access the application through portal access and see only the My Appointments (Patient) report, which is filtered so that each patient can view only records belonging to them.

Staff users can add and edit appointments, doctor and patient profiles, but cannot delete doctors. The administrator has full access. Patients are given portal access only; the portal exposes no forms other than reports permitted by the profile, and the report criteria restrict the rows to the logged-in patient.

5.9 AI and Analytics Features

Analytics are provided by the dashboard's KPI cards and two bar charts. Zia and other AI features were not used: with only a handful of appointments, predictions would add little value. All scripts were written and verified against the live application.

5.10 Pro-Code Extension

No Catalyst function, widget or Creator REST API extension was needed. If the clinic grows, Catalyst could host heavy processing such as automated billing.

5.11 Mobile and Portal Access

Patients use the customer portal to open My Appointments (Patient). The same application is available through the Zoho Creator mobile app without extra development.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing covered unit testing of each form and script, integration testing of the workflows through full booking-to-completion scenarios, role-based access testing by logging in as staff and as a portal patient, and user acceptance testing.

- Unit testing: each form was submitted with valid and invalid values to check mandatory fields and lookups.
- Workflow testing: each Deluge script was triggered by selecting a doctor, opening a new form and completing an appointment.
- Report and dashboard testing: counts on the dashboard were compared with the rows in the reports.
- Access testing: the portal report was opened as a patient to confirm only own records appear.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Expected Result	Status
TC-01	FR-01	Add a doctor with all mandatory fields	Doctor appears in All Doctor Profiles	Pass
TC-02	FR-02	Add a patient with all mandatory fields	Patient appears in All Patient Profiles	Pass
TC-03	FR-03	Book appointment with all fields	Record saved and listed	Pass
TC-04	FR-03	Submit appointment with Date empty	Validation error; not saved	[Record]
TC-05	FR-04	Open new appointment form	Status shows Scheduled	[Record]
TC-06	FR-05	Select a doctor	Consultation fee auto-filled	[Record]
TC-07	FR-06	Change status to Completed	Completion Date = today (29-Oct-2026 observed)	Pass
TC-08	FR-07	Set status to Cancelled	Record remains with blank Completion Date	Pass
TC-09	FR-08	Open Dashboard	KPI cards show 4 doctors, 3 patients, 2 appointments	Pass
TC-10	FR-08	Compare status summary with report	1 Completed, 1 Cancelled, 0 Scheduled	Pass
TC-11	FR-09	Open Statuses report	Records grouped by status	Pass
TC-12	FR-09	Open Specializations and Genders	Groups displayed correctly	Pass
TC-13	FR-10	Patient opens My Appointments (Patient)	Only own appointments visible	[Record]
TC-14	NFR-02	Patient tries to open Dashboard	Access denied	[Record]
TC-15	FR-11	Click Quick Link buttons	Respective form / report opens	[Record]
TC-16	FR-03	Book appointment with a patient not yet registered	Patient must be registered first; lookup lists only existing patients	[Record]
TC-17	FR-05	Change the doctor on an open form	Fee updates to the new doctor's fee	[Record]
TC-18	FR-06	Edit a Completed record without changing status	Completion Date is not overwritten	[Record]
TC-19	FR-09	Add a new doctor and open Specializations	New doctor appears in its group	[Record]
TC-20	NFR-06	Cancel an appointment and open All Appointment Records	Cancelled record still listed	Pass

6.3 Debugging Deluge Scripts

Scripts were debugged with info statements printed in the Deluge editor, running them against test records, and reading the error messages shown on form submission. A typical issue was a field link name mismatch (for example Consultation_Fee) that left the fee blank; correcting the link name fixed it.

Debugging also involved checking execution order: because DS-01 runs on load and DS-03 on edit, the two scripts never act on the same event, which avoids one script overwriting the other.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was developed and tested in the Development environment and published as a live application Version notes were recorded at each publish, and the previous version can be restored if a release fails.

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done
1	All test cases executed	[Yes]
2	Test data reviewed or removed	[Yes]
3	Roles, permissions and portal users configured	[Yes]
4	Workflows enabled	[Yes]
5	Application published and URL verified	[Yes]

Before each publish, the developer re-checked the lookup targets, the field link names used in the Deluge scripts and the report criteria, because a renamed field would silently break a script. Version notes recorded what changed in each release.

7.2 Performance Optimisation

Scripts fetch a single record by its ID rather than looping, reports show only the columns needed, and the patient portal report is filtered by criteria so it loads only the patient's own records. At the current data volume, response times are immediate, so no before/after measurement was necessary.

7.3 Security Features

Table 7.2: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role / profile permissions	Forms, reports and page	Unauthorised data access
Record-level criteria	My Appointments (Patient)	Patient viewing others' records
Portal access	Patient login	Exposure of staff-only screens
No stored credentials	No external connections	Credential leakage

7.4 Governance and Administration

Governance relies on Creator's audit trail and record history, consistent naming of forms and fields, retention of cancelled records, and periodic export of data as backup. The administrator manages user licences and is responsible for maintenance.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The scenario below follows two real appointments through the system. Patient joseph booked Dr. Yashu for 14-Oct-2026 at 11:00 (“not well”); the appointment was later cancelled but stays in the database for history. Patient ramya booked Dr. Priya for 29-Oct-2026 at 10:30 (“not feeling well”), scheduled by user san; it was marked Completed, the system filled Completion Date 29-Oct-2026 and Reminder Sent shows true.

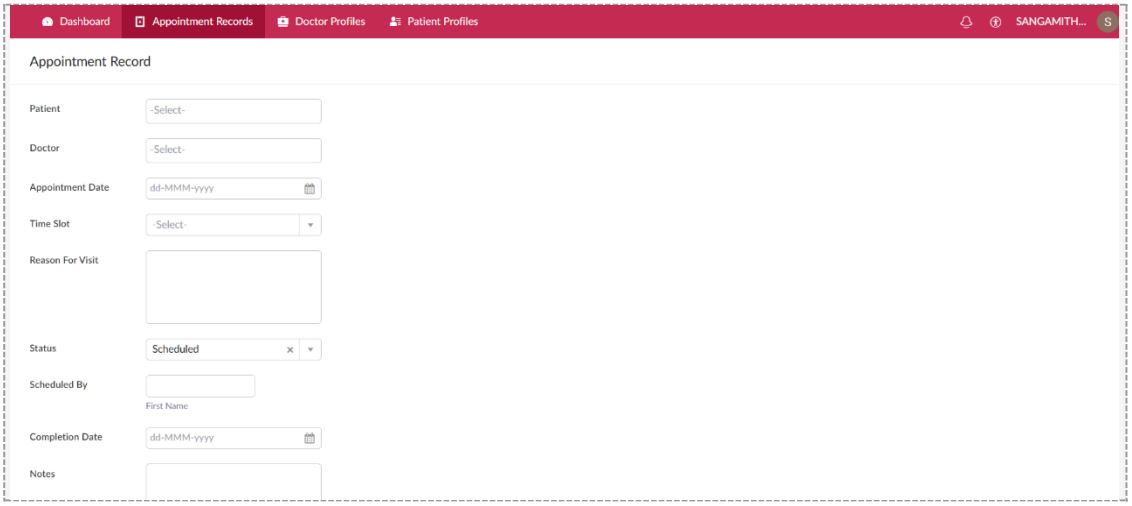


Figure 8.1: Scenario Step 1 – Booking an Appointment

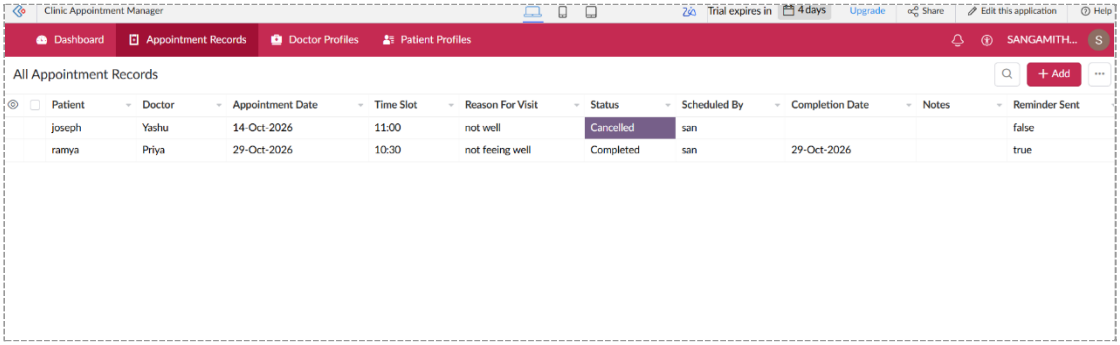


Figure 8.2: Scenario Step 2 – Appointment Marked Completed

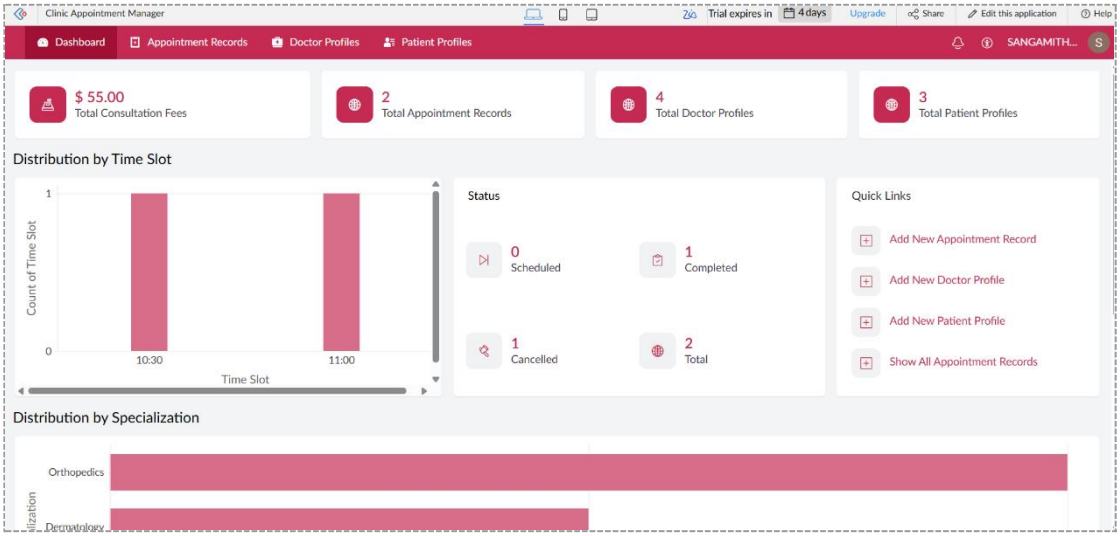


Figure 8.3: Scenario Step 3 – Dashboard Updated

Table 8.1: Live Appointment Records Used in the Walkthrough

Patient	Doctor	Appointment Date	Time Slot	Reason For Visit	Status	Completion Date	Reminder Sent
joseph	Yashu	14-Oct-2026	11:00	not well	Cancelled	(blank)	false
ramya	Priya	29-Oct-2026	10:30	not feeling well	Completed	29-Oct-2026	true

Both records were scheduled by the staff user san. The totals shown on the dashboard (two appointments, $55.00 in fees) agree with these records, which confirms that the reports and dashboard read from the same data.

The workflow evidence is therefore twofold: the automatic completion date on the Completed record, and the preserved Cancelled record, which shows that no history is lost when an appointment is cancelled.

8.2 Achievement of Objectives

Table 8.2: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Section	Outcome / Evidence
Data model and forms	4.2, 5.2	3 linked forms with lookups
Deluge automation	5.5, 5.6	Default status, fee lookup, completion date observed
Dashboard and reports	5.3, 5.4	7 reports; dashboard with 4 KPIs, 2 charts
Patient portal	5.8	Filtered My Appointments (Patient)
Secure deployment	7.1–7.4	Roles, portal and governance in place

8.3 Comparison with the Existing System

Table 8.3: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Fee entry	Manual	Auto-filled from doctor	Fewer errors
Completion date	Manual note	Auto-stamped	Consistent history
Reporting	Manual counting	Live dashboard and grouped reports	Real-time
Patient access	Phone call	Portal report	Self-service

8.4 Limitations of the Low-Code Approach

The dashboard layout is limited to the page components the platform provides, Deluge has execution and API-call limits, and the application depends on the Zoho platform (vendor lock-in). None of these affected this clinic-scale application.

Another practical limitation is that the application's look is bound by the platform's themes and page components. This suited a clinic scheduler, where clarity matters more than custom styling, but a clinic wanting a branded patient-facing app might need a pro-code front end on top of the Creator data.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Forms, reports, lookups, grouping and the dashboard were pure configuration (no-code). Default status, fee lookup and completion date needed short Deluge scripts (low-code). Payment processing, SMS gateways or heavy analytics would call for pro-code extensions such as Catalyst.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Clinic Appointment Manager replaces a manual appointment register with three linked forms, seven reports, a live dashboard and three Deluge automations, plus a patient portal view. The live data shows the automations and reports working as designed, and the objectives set in Chapter 1 were met on the Zoho Creator platform.

The system therefore meets its purpose of giving a small clinic structured records, automatic rules and a live overview, and it provides a base on which the enhancements below can be built.

9.2 Future Enhancements

- E-mail or SMS reminders through a scheduled workflow (the Reminder Sent flag is already in place).
- Doctor-wise availability and conflict checking for time slots.
- Payment gateway integration for consultation fees.
- Predictive analytics with Zia and a Catalyst service for heavy processing.

The project also showed that careful data modelling matters more than code: once the three forms and their lookups were right, reports, grouping and the dashboard followed with very little effort.

9.3 Learning Outcomes and Career Relevance

The project built skills in data modelling, Deluge scripting, dashboards and access control, which map to roles such as low-code developer, citizen developer and automation engineer.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Section
I	Application created & sections / menu organised	Y	4 navigation tabs	5.1
II	Forms with ≥ 8 distinct field types	Y	Appointment Record	5.2
II	Lookup field	Y	Patient, Doctor	4.2, 5.2
II	Subform	N	Not required	—
II	Field validations / layout rules	Y	Mandatory fields	5.2
II	List report with filter and grouping	Y	All Appointment Records, Statuses	5.3
II	Kanban / Calendar / Timeline / Map report	N	Not used	—
II	Pivot table / chart report	Y	Dashboard bar charts	5.4
II	Page / dashboard built with UI Builder	Y	Dashboard	5.4
II	Deluge – form events	Y	DS-01, DS-02, DS-03	5.5
II	Deluge – functions, fetch, loops, collections	Y	Fetch used (DS-02); no loops	5.5
II	Email / in-app notifications	N	Not used	—
II	Form workflow	Y	3 form workflows	5.6
II	Scheduled workflow	N	Future work	—
II	Approval process / Blueprint / Batch	N	Not required	—
II	Testing & debugging evidence	Y	Table 6.1	6
III	Zoho service integration / REST API / Webhook	N	Not used	5.7
III	Users, roles and permission profiles	Y	Admin, Staff, Patient	5.8
III	Record-level access / customer portal	Y	My Appointments (Patient)	5.8
III	Zia / AI feature	N	Not used	5.9
III	Analytics (charts)	Y	Dashboard	5.4, 5.9
IV	Performance optimisation applied	Y	ID-based fetch, filtered report	7.2
IV	Deployment across environments	Y	[confirm]	7.1
IV	Security controls	Y	Roles, portal criteria	7.3
IV	Governance (audit trail, administration)	Y	Audit trail, cancelled records kept	7.4
V	Pro-code extension	N	Not used	5.10
V	Low-code limitations & spectrum analysis	Y	Reflection	8.4, 8.5

APPENDIX B – REPORTS AND DASHBOARD CATALOGUE

B.1 Reports

- All Appointment Records, Statuses, My Appointments (Patient)
- All Doctor Profiles, Specializations
- All Patient Profiles, Genders

B.2 KPI Cards and Status Summary

- Total Consultation Fees ($55.00); Total Appointment Records (2); Total Doctor Profiles (4); Total Patient Profiles (3)
- Status summary: 0 Scheduled, 1 Completed, 1 Cancelled, 2 Total Appointments

B.3 Charts and Quick Links

- Distribution by Time Slot (bar chart: 10:30 and 11:00)
- Distribution by Specialization (horizontal bar chart: Orthopedics, Dermatology, Cardiology)
- Quick Links: Add New Appointment Record, Add New Doctor Profile, Add New Patient Profile, Show All Appointment Records

B.4 Recommended Monitoring

Table B.1: Recommended Monitoring

Frequency	Monitoring Activity
Daily	Review Scheduled appointments for the day; update statuses of visits that have taken place.
Weekly	Review the status mix, cancelled appointments and time-slot load.
Monthly	Review specialization demand, total consultation fees and patient registrations.

No scheduled workflow exists in the current version, so these reviews are manual.

APPENDIX C – COMPLETE DELUGE SOURCE CODE

Code C.1: DS-01 – Default status

DS-01 (Default status): Sets the appointment status to "Scheduled" immediately upon loading the "Appointment_Record" form (On Load (Add) event), ensuring all new entries start with a consistent default state.

```
// Form: Appointment_Record | Event: On Load (Add)
input.Status = "Scheduled";
```

Code C.2: DS-02 – Dynamic fee lookup

DS-02 (Dynamic fee lookup): Triggers when a user selects or changes a doctor in the "Appointment_Record" form (On User Input event). It queries the "Doctor_Profile" database using the selected doctor's ID and automatically populates the "Consultation_Fee" field with that doctor's specific rate.

```
// Form: Appointment_Record | Field: Doctor | On User Input
if(input.Doctor != null)
{
    doc = Doctor_Profile[ID == input.Doctor];
    input.Consultation_Fee = doc.Consultation_Fee;
}
```

Code C.3: DS-03 – Auto completion date

DS-03 (Auto completion date): Executes during the validation phase when editing an appointment record (On Edit - On Validate event). If the status is updated to "Completed" and the completion date field is currently empty, it automatically assigns the current system date (zoho.currentdate) to the "Completion_Date" field.

```
// Form: Appointment_Record | Event: On Edit - On Validate
if(input.Status == "Completed" && input.Completion_Date == null)
{
    input.Completion_Date = zoho.currentdate;
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the application URL [https://creatorapp.zoho.in/...] and sign in with your Zoho account. Staff and administrators see the Dashboard first; patients are taken to the portal.

D.2 Instructions for Staff

- Register the patient under Patient Profiles → + Patient Profile; doctors under Doctor Profiles → + Doctor Profile.
- Book a visit under Appointment Records → + Appointment Record: choose patient, doctor, date and time slot, then enter the reason for visit. The fee and status fill automatically.
- After the visit, edit the record and set Status to Completed; the Completion Date is filled automatically. For a cancellation, set Status to Cancelled.
- Use Statuses, Specializations and Genders to view grouped data, and the Dashboard for a live summary.

D.3 Instructions for Patients

- Sign in to the portal and open My Appointments (Patient) to see the date, time slot, doctor and status of your own appointments.

D.4 Screen Reference

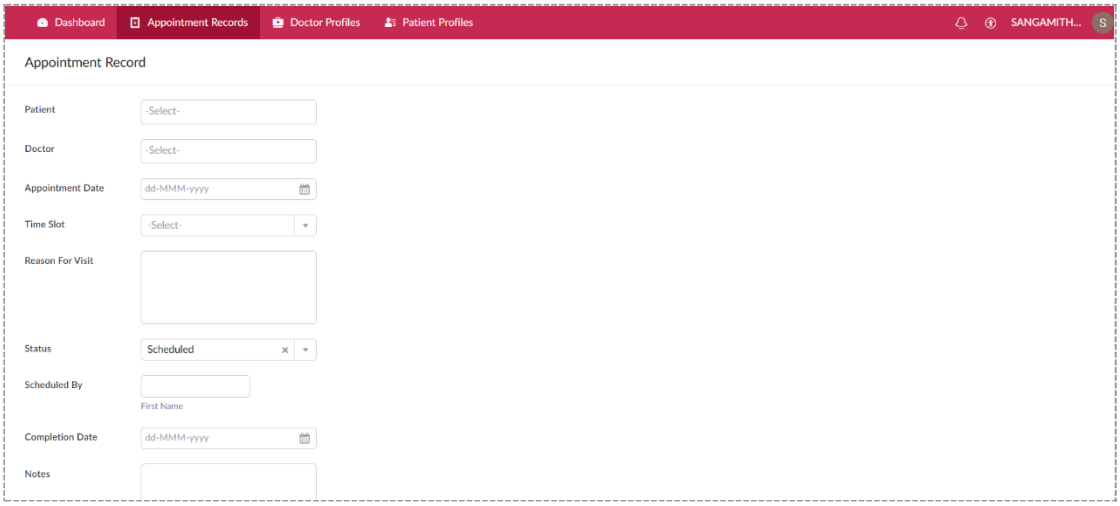


Figure D.1: Booking Screen Reference

D.5 Troubleshooting

Table D.1: Troubleshooting

Issue	Recommended Check
Fee is blank on the appointment form	Confirm that a doctor is selected and that the doctor profile has a consultation fee.
Status is not Scheduled on a new record	Check that the on-load script for the Appointment Record form is enabled.
Completion Date is empty for a Completed record	Confirm Status is set to Completed and the record was saved in edit mode.
Patient sees no appointments	Confirm the patient has portal access and that appointments exist for that patient.
Dashboard totals look wrong	Compare the counts with All Appointment Records and refresh the page.

APPENDIX E – EVALUATION RUBRIC

Student Name: SANGAMITHRA ARUN Register No.: 240701465

Project Title: Clinic Appointment Manager Date:

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	% of Max.	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner: Signature of External Examiner:

_____ _____

APPENDIX F – VIVA-VOCE RECORD SHEET

Student Name	SANGAMITHRA ARUN	Register No.	240701465
Department	Computer Science and Engineering	Date of Viva	
Application URL Verified	Yes /No	Demo Duration	_____min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E/G/S/NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners' Remarks:_____

INTERNAL EXAMINER EXTERNAL EXAMINER

Name: Name:

: